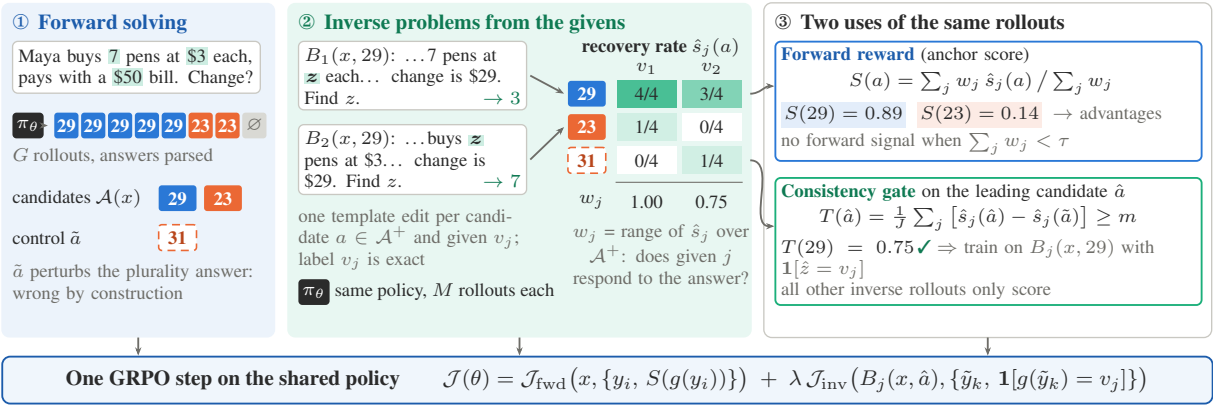

① Forward solving
Maya buys 7 pens at $3 each, pays with a $50 bill. Change?
π_θ 29 29 29 29 29 23 23 ∅
G rollouts, answers parsed
candidates $\mathcal{A}(x)$ 29 23
control $\tilde{a}$ 31
$\tilde{a}$ perturbs the plurality answer: wrong by construction
② Inverse problems from the givens
$B_1(x, 29)$: …7 pens at z each… change is $29. Find z. → 3
$B_2(x, 29)$: …buys z pens at $3… change is $29. Find z. → 7
one template edit per candidate $a \in \mathcal{A}^+$ and given v_j; label v_j is exact
π_θ same policy, M rollouts each
recovery rate $\hat{s}_j(a)$
v_1 v_2
29 4/4 3/4
23 1/4 0/4
31 0/4 1/4
w_j 1.00 0.75
w_j = range of $\hat{s}_j$ over $\mathcal{A}^+$: does given j respond to the answer?
③ Two uses of the same rollouts
Forward reward (anchor score)
$S(a) = \sum_j w_j \hat{s}_j(a) / \sum_j w_j$
$S(29) = 0.89$ $S(23) = 0.14$ → advantages
no forward signal when $\sum_j w_j < \tau$
Consistency gate on the leading candidate $\hat{a}$
$T(\hat{a}) = \frac{1}{J} \sum_j [\hat{s}_j(\hat{a}) - \hat{s}_j(\tilde{a})] \geq m$
$T(29) = 0.75$ ✓ ⇒ train on $B_j(x, 29)$ with $\mathbf{1}[\hat{z} = v_j]$
all other inverse rollouts only score
One GRPO step on the shared policy
$\mathcal{J}(\theta) = \mathcal{J}_{\text{fwd}}(x, \{y_i, S(g(y_i))\}) + \lambda \mathcal{J}_{\text{inv}}(B_j(x, \hat{a}), \{\tilde{y}_k, \mathbf{1}[g(\tilde{y}_k) = v_j]\})$